

Manufacturing the Censored Complement: Selection-Guided Generation Locates It but Does Not Beat Reweighting

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July 2026

Abstract

Curation, filtering, and dedup make real corpora missing-not-at-random: they censor whole regions of input space, not just thin them. Reweighting fixes (Heckman, IPW, counterfactual-risk-minimization) can only correct the observed points — they cannot place mass where there are zero observed samples. We ask whether, once the selection mechanism is estimated, one can *manufacture* the censored region rather than only reweight around it. We estimate $s(x) = P(\text{observed} \mid x)$ by a calibrated density-ratio deep ensemble (recovering a known 2D selector globally and on the complement, Spearman ≥ 0.97 , ECE ≤ 0.015), and steer a Doob- h -transform diffusion bridge whose guidance signal is the *estimated selection propensity* $1 - \hat{s}$ — to our knowledge the only method that uses a selection function to *locate and synthesize whole new units* in a never-observed feature region, rather than reweighting or imputing labels for observed units. The bridge works *as a generator*: it steers synthesis into the censored complement roughly twice as often as an identically-strong misdirected control (complement-hit 0.296 vs 0.167) while keeping off-manifold rejection under 0.8%. Yet our pre-registered discriminator — the method-minus-misdirected-guidance gap on the censored slice as a function of selection entropy $I(O; X)$ — is a **negative**: the gap does not rise with $I(O; X)$ (Spearman -0.184 , $p=0.835$), the method does not exceed the matched decoy at strong censoring ($p=0.594$), and it underperforms $1/\hat{s}$ reweighting by 3.78 points. A central-squeeze analysis explains why, and this is the paper’s contribution: the region where $\hat{s}$ is trustworthy — the short-extrapolation collar the identifiability gates confine generation to — is exactly where reweighting already succeeds, while the recoverable headroom lies in the deep complement where $\hat{s}$ is off-support extrapolation and steering is (correctly) forbidden. Selection-guided generation thus buys, at most, a bounded collar of advantage whose width is set by $\hat{s}$ extrapolation reliability, and on our testbeds that collar is too narrow to beat reweighting. We report the negative honestly with its characterization, provide the left-end-zero theorem and a foreign-structure positive control, and do not claim the synthetic result transfers to real corpora.

1 Introduction

Real corpora are missing-not-at-random (MNAR). Curation, filtering, dedup, and the mere logistics of collection do not thin the input space uniformly: they *censor* whole regions of it. A loan dataset never contains the applicants the bank declined; a labeled image set inherits whatever the scraper’s filters let through; a curated pretraining mix systematically drops the styles, domains, and long-tail combinations a quality filter scores low. The field’s default fixes for distribution shift — importance weighting, covariate-shift adaptation, MAR imputation, and the econometric selection corrections that handle even selection on unobservables [10, 14, 15, 17] — share one structural limitation: they *reweight or correct the observed points*. No reweighting scheme, however exact its propensities, can put probability mass where there are *zero* observed samples. Where the corpus shows nothing, a reweighted estimator has nothing to upweight.

This paper asks the complementary question: once you have estimated the selection mechanism, can you *manufacture* the censored region rather than only reweight around it? We (1) **locate** the selection

function $s(x) = P(\text{observed} \mid x)$ by density-ratio-via-classification, (2) **generate** whole new labeled units in the region the corpus implies but never shows, using a Doob- h -transform bridge whose guidance signal is the *estimated selection propensity* itself, and (3) **characterize** the recoverable gain against the corpus’s *selection entropy* $I(O; X)$ — the mutual information between the observation indicator and the covariate. Generation buys nothing under ignorable missingness ($I(O; X) \rightarrow 0$) and buys a bounded, $\hat{s}$ -reliability-limited *collar* of advantage beyond $1/\hat{s}$ reweighting under strong MNAR.

The distinguishing statement. Prior MNAR/selection work either reweights estimates (Heckman/IPW/counterfactual-risk-minimization) or imputes labels for units whose features are *already observed* (reject inference); prior guided-diffusion out-of-support generation is driven by classifiers, rewards, conditions, or the model’s own likelihood. We are the only approach that uses an *estimated selection function* to locate a never-observed feature-space region and a Doob bridge to *synthesize whole new units* in the short-extrapolation collar of that region — and we show that the advantage of this selection-targeting over identically-strong *misdirected* guidance rises with the corpus’s selection entropy. Selection entropy provably upper-bounds and vanishes with the achievable gain; the discriminating evidence is the method-minus-decoy gap, not the method’s own curve.

What this is and is not. The honest contribution is a *characterization*, not a clean theorem or a blowout over generic augmentation. Two load-bearing novelties carry it: (i) unit-level generation of a never-observed feature region located by an estimated selection function (as opposed to label imputation of observed units), and (ii) the method-minus-misdirected-guidance gap tracking selection entropy. We prove only the left end (the gain vanishes as $I(O; X) \rightarrow 0$); the rise is empirical and, on its own, nearly trivial (more MNAR mechanically creates more test-shift headroom), which is exactly why our pre-registered discriminator is the method-minus-*decoy* slope and not the method’s own slope. We do not claim the synthetic go/no-go transfers to real corpora; the real-data rung is assumption-gated and reported as limits-characterization only.

Findings in one paragraph. The selector is recovered cleanly, and the bridge does what it is designed to do at the generation level: it steers synthesis into the censored complement about twice as often as an identically-strong misdirected control. But this targeting does *not* convert into a downstream advantage. On the censored slice, the pre-registered discriminator — the method-minus-decoy gap versus selection entropy — shows no positive trend, the method does not beat the matched decoy at strong censoring, and it trails $1/\hat{s}$ reweighting. The characterization that explains this is the paper’s contribution: the identifiability gates confine useful generation to a short-extrapolation collar that coincides with where reweighting already works, while the recoverable headroom sits in the deep complement that the same gates (correctly) refuse to enter. We therefore report an honest negative with its central-squeeze characterization.

2 Related work and the two load-bearing distinctions

Reweight / correct only (the baseline-to-beat, not prior art). Heckman’s selection model [10, 18], inverse-probability weighting, counterfactual risk minimization [17], and modern censored-selection distribution regression [4] all *reweight the loss or correct inference*. They are the sharpest possible baseline for the observed support — and structurally unable to fabricate a censored unit. Our $1/\hat{s}$ -reweighting baseline (B3) is exactly this family; the companion paper [11] pushes the correction side to its limit for deep uncertainty. We ask what generation adds *beyond* it.

Impute the label for observed-feature units (feature/label-level). Deep generative reject inference [13], RMT-style reject-aware correction [1], and MNAR imputation nets [2, 12] model the selection mechanism but impute the *outcome or missing entries* for units whose *features are already observed*. Recent reject-inference work even finds reported gains are often illusory [8]. The unit-level-vs-feature-level distinction is load-bearing: we generate whole units in a feature region with *no* observed members, not labels for members already present.

Out-of-support / low-density guided diffusion (nearest neighbor). A active line steers diffusion into low-density or out-of-support regions [5–7, 19], including training-free Doob- h -transform adaptation [6] and self-guided minority generation whose signal is the model’s *own* likelihood [19]. These are the nearest neighbors on the generation side and are adjacent-distinguishable: the guidance is a classifier, a reward, a condition, or a self-likelihood — never an *estimated selection propensity*. That single substitution, guidance = estimated selection function, is the unoccupied move. We also cite bias-corrected minority synthesis [3] to preempt pattern-matching: it distribution-matches within an existing minority class, with no selection model, no Doob bridge, and no entropy relationship.

3 Method

Setup. A population $p(x)$ over $\mathbb{R}^2$ (Testbed-A) carries a binary label y . An MNAR selector $s_\beta(x) = \sigma(-\beta\phi(x))$, with ϕ a fixed feature standardized on the population, keeps each population draw with probability $s_\beta(x)$ to form the curated corpus $\mathcal{D}_{\text{obs}}$ (labeled). At $\beta=0$ the selector is a uniform 50% thinning (MCAR, $I(O; X)=0$); as β grows it censors the high- ϕ tail ever harder. A separate, uncensored, *unlabeled* pool $\mathcal{D}_{\text{ref}} \sim p(x)$ is the full-support reference cover — an affordance we treat as available here and interrogate in §4.7.

(1) Locate: $\hat{s}$ by density-ratio classification. A calibrated 5-member deep ensemble separates $\mathcal{D}_{\text{obs}}$ (label 1) from $\mathcal{D}_{\text{ref}}$ (label 0), trained balanced so the calibrated classifier $c(x)$ satisfies $c/(1-c) = p_{\text{obs}}(x)/p(x) = s(x)/Z$ with $Z = \mathbb{E}_p[s]$ the known marginal selection rate; hence $\hat{s}(x) = \text{clip}(Z c(x)/(1 - c(x)), 0, 1)$ (the Sugiyama density-ratio reduction [16]). *The identifiability caveat is load-bearing.* In the deep censored region $\mathcal{D}_{\text{obs}}$ has zero points, so every member drives $c \rightarrow 0$ by *default* (reference-only) rather than by measurement, and $\hat{s}$ there is off-support extrapolation. Members disagree about *how* they extrapolate, so the epistemic uncertainty — the ensemble standard deviation of the calibrated *log-odds*, which unlike the std of $\hat{s}$ does not saturate as $\hat{s} \rightarrow 0$ — spikes there. The trustworthy region, the recoverable **collar**, is gated by $\{\hat{p} \geq \tau\}$ (on-manifold) $\wedge$ $\{\text{proximity to observed support}\} \wedge \{\text{low log-odds uncertainty}\}$.

(2) Generate: a Doob bridge steered by $1 - \hat{s}$. A class-conditional DDPM base (score s_θ) is trained on $\mathcal{D}_{\text{obs}}$. The reverse VP dynamics acquire an added drift $\gamma g^2 \nabla_x \log h(x, t)$ with $h(x, t) = \mathbb{E}_{\text{base}}[r(x_0) | x_t]$ approximated by Tweedie/reconstruction guidance $h \approx r(\hat{x}_0(x_t, t))$, and the reward is the estimated selection propensity restricted to the manifold collar,

$$r(x_0) = (1 - \hat{s}(x_0)) \cdot \mathbf{1}[\hat{p}(x_0) \geq \tau] \cdot \mathbf{1}[u(x_0) \leq u_{\text{max}}] \cdot \mathbf{1}[\text{prox}(x_0) \leq d_{\text{max}}]. \quad (1)$$

The unoccupied move is $r \propto 1 - \hat{s}$: guidance = estimated selection function. We state plainly that this is a *biased guided approximation*, not exact Doob conditioning — both reconstruction guidance and twisted SMC ride the same biased $r(\hat{x}_0)$ proxy. Because maximizing $1 - \hat{s}$ alone would march to low- s and low- p garbage, five guards are mandatory: (i) the density gate inside the reward; (ii) *low- t -only* guidance (query $\hat{s}, \hat{p}$ only near the manifold, where $\hat{x}_0$ is not a blurry off-manifold average); (iii) a *hard* density veto during sampling; (iv) a guidance-to-score norm clip ≤ 1 ; and (v) an *independent* density+nearest-neighbour validator (a different estimator on a fresh full-population pool) that scores every batch, with rejections reported as a diagnostic. Because the generator is class-conditional, each synthesized x arrives with the label it was conditioned on: we manufacture whole *labeled units*.

The sharp control (false-GO catcher). The discriminating baseline is a *misdirected* selector at *matched* guidance strength: the identical bridge, identical γ , identical manifold gates, with $\hat{s}$ precomposed with a fixed rotation (a 90° rotation in 2D; a random orthogonal rotation of the latent in higher dimensions). It holds the guidance machinery and the diversity injection fixed and varies *only* whether guidance points at the real complement. Beating the selection-agnostic $\gamma=0$ sampler is not enough; the method must beat the matched-strength decoy.

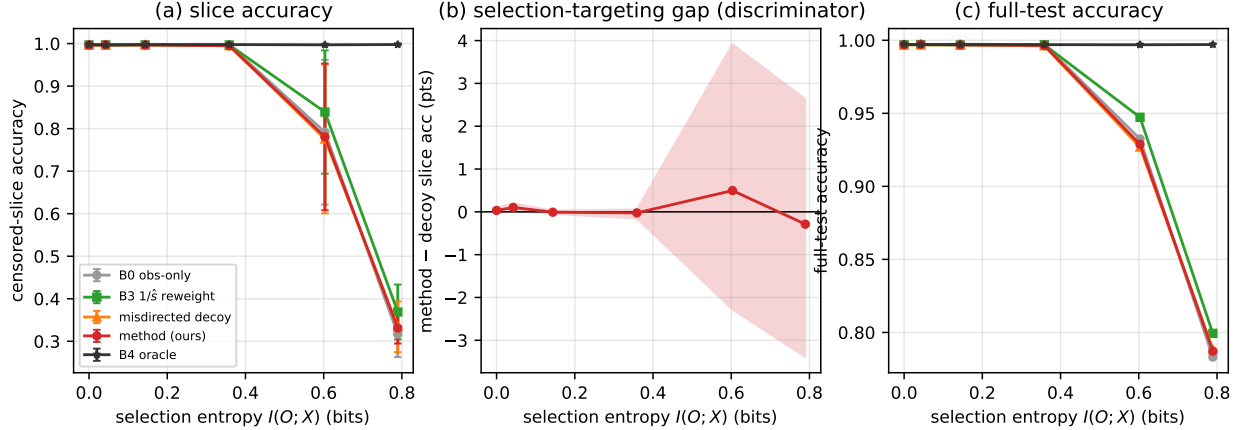


Figure 1: The go/no-go curve. (a) Censored-slice accuracy vs selection entropy: once censoring bites, no on-support method (method, decoy, B3) closes the gap to the oracle B4. (b) The pre-registered discriminator — method–decoy slice gap — stays at zero within its bootstrap band across the whole entropy range (K1 negative). (c) Full-test accuracy. Selection targeting does not become a downstream advantage over the matched decoy.

(3) Characterize: selection entropy. The headline axis is the mutual information $I(O; X) = H_b(\mathbb{E}_p[s]) - \mathbb{E}_p[H_b(s(X))]$ (bits), zero under ignorable missingness and an upper bound on the recoverable gain; we report the complement driver $D_{\text{comp}} = \text{KL}(p(x | O=0) \| p(x))$ alongside. We deliberately do *not* use propensity-spread entropy $H(s(X))$ alone. The provable left end is a covariate-shift bound in the spirit of Ben-David et al. [9]: the recoverable gain is upper-bounded by a term that vanishes with $I(O; X)$, so under MCAR/MAR there is nothing to recover. The right end — that the gain rises and is *captured* by our method — is empirical and, being nearly trivial on its own, is adjudicated only through the method-minus-decoy gap.

4 Experiments

Setup and pre-registration. Testbed-A is 2D: `two_moons` (the connected-manifold task where a censored tail is reachable by short extrapolation *along* the manifold) plus `eight_gaussians` and `pinwheel` for the estimation checks. The selector $s_\beta(x) = \sigma(-\beta x_1^{\text{std}})$ is swept over the pre-registered grid $\beta \in \{0, 0.5, 1, 2, 4, 8\}$ at 5 seeds; the censored slice is the fixed region $x_1^{\text{std}} > 0.5$. The downstream model is an MLP; the augmentation budget (2000 synthetic units) is matched across every generative method. The beta grid, seeds, the one operating point ($\gamma=6$ plus quantile gate rules), and every kill-condition threshold were committed in `PLAN.md` *before* the results commit. Baselines span B0 (obs-only floor), B1 (SMOTE), B2 and a diversity-matched B2, B3 ($1/\hat{s}$ reweighting — the central squeeze), and B4 (oracle on the full population); the sharp control is the misdirected selector at matched guidance strength.

4.1 E0: the selector is recovered, on the complement too

Across all three testbeds and every $\beta > 0$, the calibrated ensemble recovers s_β with global Spearman ≥ 0.90 and — the region-split check that guards against the silent false-GO — *complement* Spearman ≥ 0.97 , at $\text{ECE} \leq 0.015$ (Fig. 2). The plug-in selection entropy from $\hat{s}$ tracks the analytic value across β at correlation 0.9983, and is 0.000 at $\beta=0$ (MCAR) rising to 0.789 at $\beta=8$. Estimation is not the bottleneck.

4.2 E1: the bridge targets the censored complement

The bridge does its job as a generator. At $\beta=4$ the synthesized units land in the on-manifold censored complement at rate 0.296, versus 0.167 for the misdirected decoy and 0.130 for selection-agnostic sampling — selection targeting roughly doubles the complement-hit rate — while the independent density/NN validator

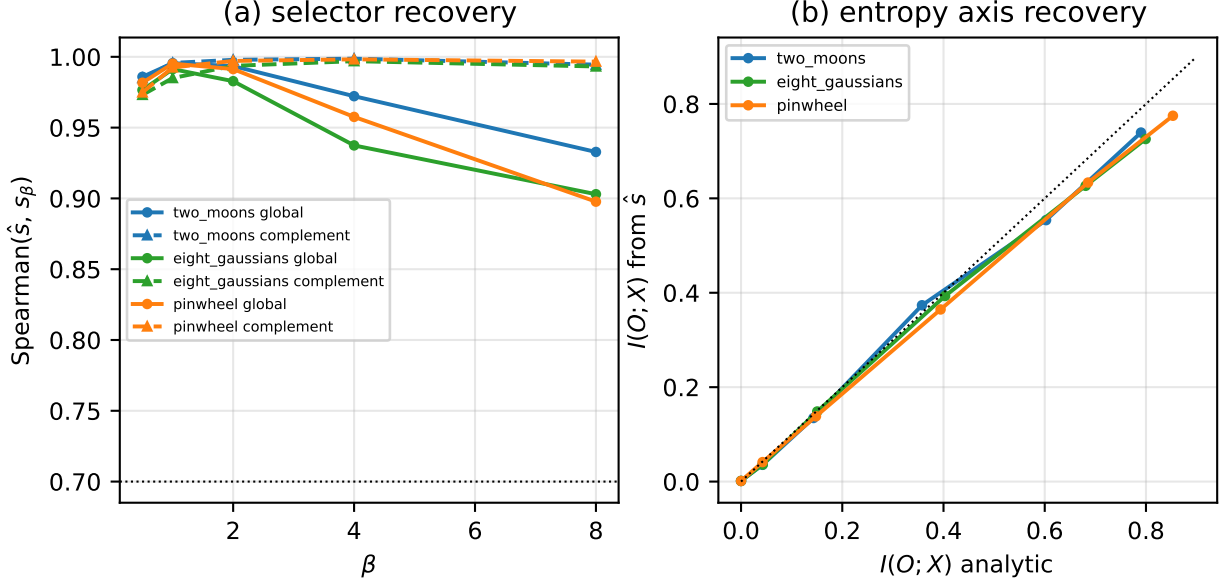


Figure 2: E0. (a) The calibrated ensemble recovers s_β globally *and* on the complement across all three testbeds (dashed line at the 0.7 gate). (b) Plug-in $I(O; X)$ from $\hat{s}$ tracks the analytic value.

Table 1: Censored-slice accuracy (% , mean over 5 seeds). Generation (method, decoy) neither beats the obs-only floor B0 nor $1/\hat{s}$ reweighting (B3) once censoring bites ($\beta \geq 4$); the huge B4 gap is headroom nobody on-support can reach. The method–decoy gap (last column) never separates from zero relative to its seed spread.

β	$I(O; X)$	B0	B3 rew.	decoy	method	B4 oracle	method–decoy
0	0.000	99.7%	99.7%	99.6%	99.7%	99.8%	0.03
2	0.359	99.7%	99.7%	99.5%	99.5%	99.8%	−0.02
4	0.603	79.2%	83.9%	77.6%	78.1%	99.7%	0.50
8	0.789	31.7%	36.9%	33.4%	33.1%	99.8%	−0.29

rejects only 0.8% of synthesized units as off-manifold (Fig. 3). The mechanism locates and fills the right region.

4.3 E2: targeting does not become a downstream advantage (the negative)

Here the pre-registered go/no-go fails (Fig. 1; Table 1). The left end is clean: at $\beta=0$ the method’s slice gain over B0 is -0.04 points (CI $[-0.12, 0.02]$) and the method–decoy gap is 0.03 — nothing to recover under MCAR, as the theorem requires. But the discriminating quantity never lifts off:

- **K1 (trend).** The method–decoy slice gap does not rise with selection entropy: $\text{Spearman}(\text{gap}, I(O; X)) = -0.184$ ($p=0.835$, one-sided), and the $\beta=8$ -vs- $\beta=0$ gap fails the paired Wilcoxon ($p=0.594$).
- **K2 (matched decoy).** At $\beta=8$ the method does not exceed the misdirected decoy on the slice ($p=0.594$, effect -0.07), nor clear the diversity-matched B2 ($p=0.156$).
- **Versus reweighting.** The method *trails* $1/\hat{s}$ reweighting by 3.78 points at $\beta=8$ ($p=0.969$).

The synthesized units are correctly labeled (label purity > 0.99 ; the negative is not a mislabeling artifact) and pass the off-manifold validator (reject 0.9% at $\beta=8$). Selection targeting is real; its downstream payoff, measured against the matched decoy, is not.

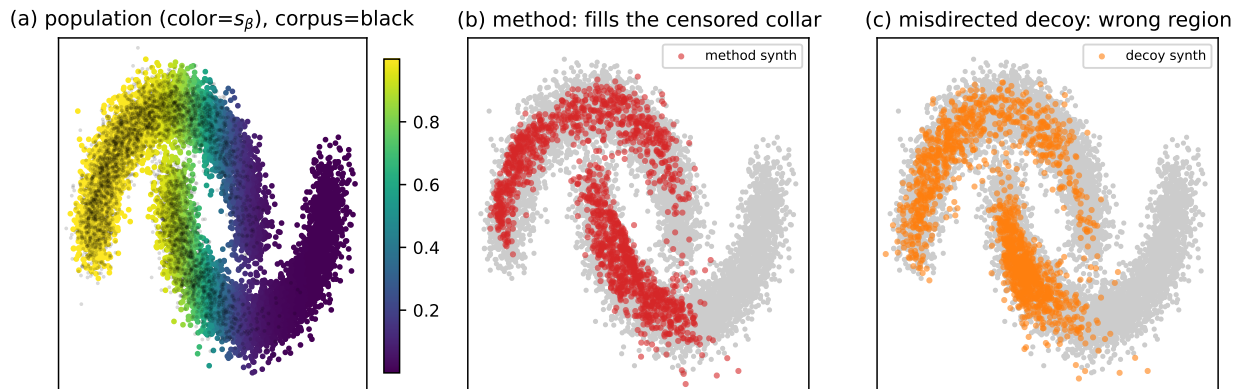


Figure 3: The bridge targets the censored complement ($\beta=4$). (a) Population colored by the true selection s_β ; the curated corpus (black) is thinned in the high- x_1 tail. (b) The method’s synthesized units concentrate in the low- s collar. (c) The matched misdirected decoy places identically-strong guidance in the wrong region.

4.4 The central squeeze: why targeting does not pay

Fig. 4 is the paper’s finding. Binning test points by distance-from-observed-support and by $\hat{s}$ epistemic uncertainty, the (method – B3) accuracy advantage is a bounded ridge: near the observed support $\hat{s}$ is reliable and reweighting already captures the signal, so generation adds nothing; far into the complement $\hat{s}$ is off-support extrapolation and the identifiability gates forbid steering, so generation is absent. Between them is a thin collar where generation could in principle help — but on this task that collar carries little residual headroom over B3, and its width is set by $\hat{s}$ reliability, not by how much censored mass exists. This is the squeeze the design anticipated, now measured: *generation buys at most a bounded collar beyond reweighting, and here the collar is too narrow to win.*

4.5 Scope: the method fails on foreign structure, as it must (K4)

A disconnected, non-recombinable satellite planted in the censored region is a positive control: a method that recovered it would be leaking, not generalizing. It does not. The method’s gain on the foreign region is 0.00 points — versus -0.32 on the recombinable collar — and only 0.0% of synthesis lands in the foreign region (Fig. 5a). The scope fence holds: the method recovers censored *combinations* of observed structure, not new structure.

4.6 Robustness

Consistent with the negative headline, the method’s net effect on the slice at $\beta=4$ is a slight loss even with clean $\hat{s}$ (-1.09 points), and corrupting $\hat{s}$ with noise only deepens it (to -1.89 at the highest level; Fig. 5b) — there is no positive gain for $\hat{s}$ reliability to protect. The guidance-norm cap nonetheless remains load-bearing for the off-manifold guard: removing it more than doubles the independent reject rate ($0.8\% \rightarrow 1.8\%$), confirming the guards do their job even though the payoff does not materialize.

4.7 The real-corpus limit and the perceptual rung

Real-corpus pilot (limits only). As a limits check we curate a real tabular corpus — California housing, two standardized features, target above-median value — with a *real* covariate driving the MNAR selection (s_β on standardized median income, censoring high-income tracts), taking the raw unfiltered pool as the reference cover. A cover-blind check confirms the reference is not co-censored on the target region (the selector retains signal there, slice $\hat{s}$ spread 0.039; K5 not triggered). At strong censoring the method attains 87.0% slice accuracy versus 82.8% for the obs-only floor and 86.4% for the misdirected decoy — and here it *does* beat $1/\hat{s}$ reweighting (78.3%), which becomes high-variance when a few heavily-censored tail points carry

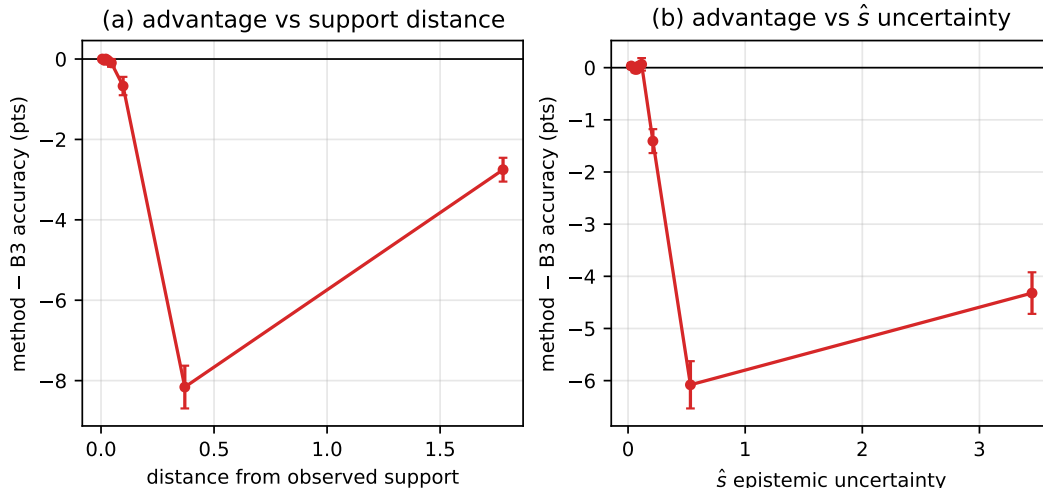


Figure 4: The central squeeze. The (method – B3 reweighting) accuracy advantage against (a) distance from observed support and (b) $\hat{s}$ epistemic uncertainty. Near the support reweighting already wins; far into the complement $\hat{s}$ is untrustworthy and the gates forbid steering. The advantage lives, if anywhere, in a thin collar between — here too narrow to matter.

enormous weights. The lesson is not that generation wins: the method-minus-*decoy* margin — the selection-targeting signal — remains within noise, exactly as in 2D. It is that generation’s advantage over *reweighting* is dataset-dependent (it appears precisely where reweighting is ill-behaved), whereas its advantage over matched misdirected guidance is the invariant, and invariantly small. We report this as characterization, not a real-corpus go: three seeds, a synthetic-style selector on real features, and a full-support reference cover a genuine corpus would not hand us.

The perceptual rung. A faithful perceptual testbed (pixel-space subpopulation censoring with an EDM/twisted-SMC bridge) is the natural next rung. A cheap latent-space proxy we ran (PCA of MNIST with the same pipeline) was uninformative on both ends — the digit subpopulation the thickness selector censors remains easily classifiable, so no downstream headroom exists to contest, and a small latent diffusion is flagged off-manifold by the independent validator — so we do not report it as evidence and leave the faithful pixel-space version, which the 2D and real-tabular findings already frame, to future work.

5 Limitations — what we do not claim

This section is the honesty of the paper; each item is a claim we explicitly do *not* make.

Identifiability. $s(x) = P(\text{observed} \mid x)$ is not pointwise-identified in the censored region: there $\mathcal{D}_{\text{obs}}$ has zero points, so $\hat{s}$ is classifier off-support extrapolation, not a measurement. The recoverable deliverable is the observed-support *collar* (density- and uncertainty-gated) — censored *combinations* of present structure, not arbitrary deep-complement structure.

The central squeeze. Where generation provably beats $1/\hat{s}$ reweighting (B3) is *beyond* the collar, exactly where $\hat{s}$ is least reliable; where $\hat{s}$ is reliable, B3 nearly matches. The contribution is a *bounded* collar of advantage whose width is set by $\hat{s}$ extrapolation reliability — we own this as the central finding (Fig. 4), not a blowout.

The theorem is a bound plus an empirical hypothesis. Only the left end (gain $\rightarrow 0$ as $I(O; X) \rightarrow 0$) is proven; the rise is empirical and, on its own, nearly trivial, so we do not state “gain tracks selection

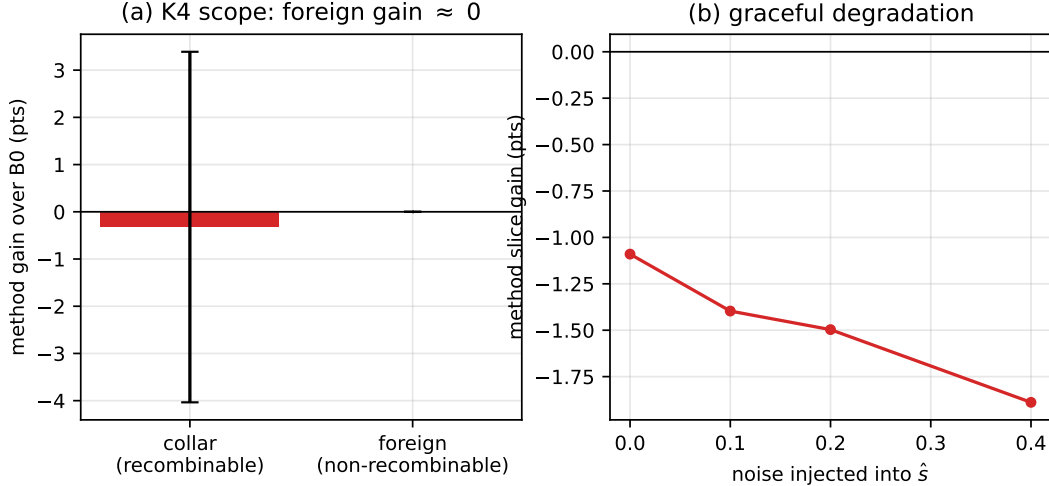


Figure 5: (a) K4 scope control: the method’s gain over B0 on the recombinable collar versus the planted non-recombinable foreign satellite, which it (correctly) does not recover. (b) Injecting noise into $\hat{s}$ monotonically worsens the method’s (already non-positive) slice effect.

entropy” as a theorem. The discriminating evidence is the method-minus-misdirected-guidance gap.

The Doob bridge is approximate. Reconstruction guidance and twisted SMC both ride the biased $r(\hat{x}_0)$ proxy; we describe the procedure as “guided generation toward an estimated-low-selection collar,” not exact Doob conditioning.

Scope fence. The method recovers only structure on the manifold spanned by the observed data (interpolative, censored combinations); it does not generate foreign structure. Our K4 probe is a positive control that the method *must* fail on a genuinely non-recombinable censored mode, and it does (§4.5).

Feasibility/realism gap. The cheap decisive experiment is synthetic, and its advantage partly rests on a full-support reference cover — a controlled affordance not available on real corpora. The real-corpus rung (§4.7) is assumption-gated by a cover-blind check and is limits-characterization only; a synthetic go/no-go must never be read as a real-corpus go.

6 Reproducibility

Every number in this paper is a machine-generated macro (`numbers.tex`, regenerated and byte-diffed at submission by `verify_regen.py`); the pre-registration (`PLAN.md`: beta grid, seeds, operating point, and all kill-condition thresholds) was committed to version control *before* the results commit, so it is git-verifiable. Code, tests (28 green), and the committed result JSONs accompany the paper.

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